RECONSTRUCTING TRIANGLE-EQUIVALENCE OF DISTANCE SETS FROM ORDERED URYSOHN AUTOMORPHISM GROUPS

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Editorial note. The native author proof is retained, including its stated extension from the displayed one-point/interleaved case to general gaps and its native TikZ figure. No individual density or maximal-subgroup lemma is separately counted.

Editorial note. The topology in the counted statement is the nonarchimedean automorphism-group topology of the relational ordered Fraisse structure. The preceding unordered theorem is only included to retain the author-referenced distance-relabeling argument; its other directions are not used as the ordered reconstruction proof.

Editorial note. Slutsky ordered partial-isometry extension is an explicit imported reference theorem; its statement is preserved in context, and its proof is not represented as locally reproduced.

Editorial note. 2026-10-11 材料收口：复用既有 H2、来源许可和近邻判断；核对原命题与作者证明起止、所指上下文及装配范围齐备。未重证作者结论；沿用既有非重复依据。

Allowed standard prerequisites (editorial).

- Fraisse limits and nonarchimedean permutation topology. Existence, uniqueness, universality and ultrahomogeneity of the Fraisse limit of finite ordered Delta-metric spaces; basic open pointwise stabilizers; open subgroups are closed; conjugacy of finite stabilizers under the action.
- Explicit classical extension and displacement inputs. Slutsky Theorem4.12 and ordered Remark4.13, exactly as quoted at653-663, provide simultaneous extensions of two order-preserving partial isometries and a free-amalgam word, with distances in the generated additive semigroup. Neumann finite-set displacement lemma applies when all subgroup orbits are infinite. These are explicit imported reference theorems; the alternating order-propagation and group reconstruction remain local obligations.

Definition 0.1 ([1])**.** Let Δ be a nonempty set of positive real numbers. We call Δ a *distance value set* if for all $x, y \in \Delta$,

$$\min\{x + y, \sup \Delta\} \in \Delta.$$

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Briefly speaking, a distance value set is exactly an initial segment of a subsemigroup of positive reals. Given a distance value set Δ, a metric space (X, d) is a Δ-*metric space* if $d(x, y) \in \Delta$ for all $x \neq y \in X$.

Fix a countable distance value set Δ. We can view each countable Δ-metric space as a countable structure, as follows. Let $\mathcal{L}_\Delta = \{R_s : s \in \Delta\}$, where each R_s is a binary relation symbol. Given any countable Δ-metric space (X, d), set $R_s^X(x, y)$ if and only if $d(x, y) = s$. Thus we turn (X, d) into a countable $\mathcal{L}_\Delta$-structure $(X; (R_s^X)_{s\in\Delta})$. Let $\mathrm{Aut}(X, d)$, or simply $\mathrm{Aut}(X)$ if d is understood, denote the automorphism group $\mathrm{Aut}(X; (R_s^X)_{s\in\Delta})$. Then $\mathrm{Aut}(X)$ is a Polish group.

Fix a countable distance value set Δ. Let $\mathcal{K}_\Delta$ be the class of all finite Δ-metric spaces. Then $\mathcal{K}_\Delta$ is a Fraïssé class. We denote the Fraïssé limit of $\mathcal{K}_\Delta$ by $\mathbb{U}_\Delta$ and call it the Δ-*Urysohn space*. In the case that $\Delta = \mathbb{Q}$, the resulting Fraïssé limit $\mathbb{U}_\mathbb{Q}$ is the well-known *rational Urysohn space* and has been denoted by $\mathbb{QU}$ in the literature. The completion of $\mathbb{QU}$ is the *universal Urysohn space* and has been denoted by $\mathbb{U}$ in the literature. $\mathbb{U}$ is universal for all separable complete metric spaces, i.e., for any separable complete metric space (X, d), there is an isometric embedding from X into $\mathbb{U}$.

To apply the KPT correspondence, we need to consider a Fraïssé ordered class. Let Δ be a countable distance value set. Let $\mathcal{K}_\Delta^<$ be the class of all finite ordered Δ-metric spaces, i.e., each element of $\mathcal{K}_\Delta^<$ is a structure $(X; (R_s^X)_{s\in\Delta}, <^X)$, where $<^X$ is a linear order on X. It is easy to see that $\mathcal{K}_\Delta^<$ is a Fraïssé (ordered) class. We denote its Fraïssé limit by $\mathbb{U}_\Delta^<$. It can be shown that $\mathbb{U}_\Delta^< \cong (\mathbb{U}_\Delta, <^{\mathbb{U}_\Delta})$ where $<^{\mathbb{U}_\Delta}$ is a linear order isomorphic to $(\mathbb{Q}, <)$ (see, e.g., Proposition 5.2 of [3]).

Definition 0.2 ([1])**.** Let Δ and Λ be distance value sets.

(i) A triple (x, y, z) of positive real numbers is a Δ-*triangle* if $x, y, z \in \Delta$ and $|x - y| \leq z \leq x + y$.
(ii) We say that Δ and Λ are *equivalent* if there is a bijection $f : \Delta \to \Lambda$ such that for any positive real numbers x, y, z, (x, y, z) is a Δ-triangle if and only if $(f(x), f(y), f(z))$ is a Λ-triangle.

Theorem 0.1 ([1])**.** *Let Δ and Λ be countable distance value sets. Then the following are equivalent:*

(i) $\mathrm{Aut}(\mathbb{U}_\Delta)$ *and* $\mathrm{Aut}(\mathbb{U}_\Lambda)$ *are isomorphic as abstract groups.*
(ii) $\mathrm{Aut}(\mathbb{U}_\Delta)$ *and* $\mathrm{Aut}(\mathbb{U}_\Lambda)$ *are isomorphic as topological groups.*
(iii) Δ *and* Λ *are equivalent.*

Theorem 0.2. *Let Δ and Λ be countable distance value sets. Then the following are equivalent:*

(i) $\mathrm{Iso}(\mathbb{U}_\Delta)$ *and* $\mathrm{Iso}(\mathbb{U}_\Lambda)$ *are isomorphic as abstract groups.*
(ii) $\mathrm{Iso}(\mathbb{U}_\Delta)$ *and* $\mathrm{Iso}(\mathbb{U}_\Lambda)$ *are isomorphic as topological groups.*
(iii) Δ *and* Λ *are equivalent.*

Proof. To see that (i) implies (iii), the key oberservation is that $\mathrm{Iso}(\mathbb{U}_\Delta)$ and $\mathrm{Aut}(\mathbb{U}_\Delta)$ are identical abstract groups for any Δ. Thus this implication follows from the implication of (iii) from (i) in Theorem 0.1.

The implication of (i) from (ii) is obvious.

To see that (iii) implies (ii), fix a bijection f from Δ to Λ which witnesses the equivalence of the two distance value sets. Let d be the metric on $\mathbb{U}_\Delta$. Define a new metric d_f on $\mathbb{U}_\Delta$ by setting $d_f(x,y) = f(d(x,y))$. Then $(\mathbb{U}_\Delta, d_f)$ is isometric to $\mathbb{U}_\Lambda$, since $(\mathbb{U}_\Delta, d_f)$ is a Fraïssé structure and its age is exactly the class of all finite Λ-metric spaces. Now the identity map from $(\mathbb{U}_\Delta, d)$ to $(\mathbb{U}_\Delta, d_f)$ induces a topological isomorphism from $\mathrm{Iso}(\mathbb{U}_\Delta)$ to $\mathrm{Iso}(\mathbb{U}_\Lambda)$. □

Theorem 0.3. *Let Δ and Λ be countable distance value sets. Then the following are equivalent:*

(i) $\mathrm{Aut}(\mathbb{U}_\Delta^<)$ *and* $\mathrm{Aut}(\mathbb{U}_\Lambda^<)$ *are isomorphic as topological groups.*
(ii) Δ *and* Λ *are equivalent.*

The rest of this subsection is devoted to a proof of Theorem 0.3. The outline of the proof follows the proof of Theorem 0.1 in [1].

The argument for the implication of (i) from (ii) is similar to the proof of Theorem 0.2.

To prove the other direction, we need some results due to Slutsky [4]. We work with the following concept from [4] in the Fraïssé theory of finite metric spaces.

Definition 0.3. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be finite metric spaces. For simplicity assume $\mathcal{A}$ is a substructure of both $\mathcal{B}$ and $\mathcal{C}$. Also assume $|\mathcal{B}| \cap |\mathcal{C}| = |\mathcal{A}|$. Define a finite metric space $\mathcal{D}$ as follows. Let $|\mathcal{D}| = |\mathcal{B}| \cup |\mathcal{C}|$. For $x, y \in |\mathcal{D}|$, define $d_\mathcal{D}(x,y)$ as follows. If $x, y \in |\mathcal{B}|$, then let $d_\mathcal{D}(x,y) = d_\mathcal{B}(x,y)$. Similarly, if $x, y \in \mathcal{C}$, then let $d_\mathcal{D}(x,y) = d_\mathcal{C}(x,y)$. If $x \in |\mathcal{B}|$ and $y \in |\mathcal{C}|$, then let

$$d_\mathcal{D}(x,y) = \begin{cases} \min\{d_\mathcal{B}(x,z) + d_\mathcal{C}(z,y) \colon z \in |\mathcal{A}|\}, & \text{if } |\mathcal{A}| \neq \varnothing, \\ \mathrm{diam}(\mathcal{B}) + \mathrm{diam}(\mathcal{C}), & \text{if } |\mathcal{A}| = \varnothing. \end{cases}$$

Then $\mathcal{D}$ is called the *free amalgam* of $\mathcal{B}$ and $\mathcal{C}$ over $\mathcal{A}$.

Since (ordered) metric spaces can be regarded as relational structures, for an (ordered) metric space $\mathcal{B}$, a substructure $\mathcal{A}$ of $\mathcal{B}$ generated by a set $A \subseteq |\mathcal{B}|$ satisfies $|\mathcal{A}| = A$. Thus we will identify $\mathcal{A}$ with A, and speak of A as a substructure. This does not cause confusion, and will simplify our notation.

Definition 0.4. Let A be a metric space and p a partial isometry on A. A point $x \in \mathrm{dom}(p)$ is called *periodic* if there is a natural number $n > 0$ such that

$$x, p(x), \ldots, p^n(x) \in \mathrm{dom}(p) \text{ and } p^n(x) = x.$$

The set of all periodic points of p is denoted by $Z(p)$. A point $x \in \mathrm{dom}(p)$ is called a *fixed point* if $p(x) = x$. The set of all fixed points of p is denoted by $F(p)$.

Lemma 0.4 ([4])**.** *Let D be a finite ordered metric space. Let p_1 and p_2 be order-preserving partial isometries of D such that $Z(p_1) = F(p_1)$ and $Z(p_2) = F(p_2)$. Suppose $F(p_1) \cap F(p_2) \neq \emptyset$. Then there exist a finite ordered metric space D' extending D, order-preserving partial isometries q_1 and q_2 of D' extending p_1, p_2 respectively, and an element $w \in \mathsf{F}(s,t)$, the free group generated by s,t, such that*

(1) $Z(q_1) = Z(p_1)$ and $Z(q_2) = Z(p_2)$;
(2) $D \subseteq \mathrm{dom}(q_1)$;
(3) $\mathrm{dom}(q_1) \cup w(q_1,q_2)(D)$ is the free amalgam of $\mathrm{dom}(q_1)$ and $w(q_1,q_2)(D)$ over $F(p_1) \cap F(p_2)$.

Moreover, the distances in D' are taken from the additive semigroup generated by the distances in D.

This result was originally proved for metric spaces as Theorem 4.12 of [4]. Nevertheless, as Slutsky remarked (Remark 4.13 of [4]), the above ordered version also holds.

We use the following notation in our further discussions. For a countable structure $\mathcal{A}$ and subset $B \subseteq |\mathcal{A}|$, we use $\mathrm{Aut}_B(\mathcal{A})$ to denote the subgroup of $\mathrm{Aut}(\mathcal{A})$ consisting of the automorphisms of $\mathcal{A}$ which fix each element of B. Note that if B is finite, then $\mathrm{Aut}_B(\mathcal{A})$ is an open subgroup of $\mathrm{Aut}(\mathcal{A})$.

Lemma 0.5. *Let Δ be a countable distance value set, and let $A, B \subseteq \mathbb{U}_\Delta^<$ be finite subsets. Suppose $A \cap B \neq \varnothing$. Then*

$$\langle \mathrm{Aut}_A(\mathbb{U}_\Delta^<), \mathrm{Aut}_B(\mathbb{U}_\Delta^<)\rangle \text{ is dense in } \mathrm{Aut}_{A\cap B}(\mathbb{U}_\Delta^<).$$

Proof. For notational simplicity, we assume $A \cap B = \{x_0\}$. Suppose $A = \{a_0, \dots, a_n\}$ and $B = \{b_0, \dots, b_m\}$. Again, for definiteness and notational simplicity, assume $m = n$ and $a_0 = b_0 = x_0 < a_1 < b_1 < \cdots < a_n < b_n$. Note that in general, $A \cap B = \{x_0, x_1, \dots, x_k\}$ where $x_0 < x_1 < \cdots < x_k$ and the rest of the points in A, B can be in the gaps between x_is including below x_0 and above x_k. For the general case, the argument is similar.

Let C be a finite subset of $\mathbb{U}_\Delta^<$ and let $p \in \mathrm{Aut}_{A\cap B}(\mathbb{U}_\Delta^<)$. It suffices to show that there is a $q \in \langle \mathrm{Aut}_A(\mathbb{U}_\Delta^<), \mathrm{Aut}_B(\mathbb{U}_\Delta^<)\rangle$ such that $p \restriction C = q \restriction C$. Without loss of generality, we may assume $A \subseteq C$ and $B \subseteq C$. Let $D = C \cup p(C)$. We find (to be explicitly defined later)

$$q = w(q_1,q_2)^{-1} g_1^{-1} \dots g_{2n}^{-1} r g_{2n} \dots g_1 w(q_1,q_2),$$

where $g_{2i-1} \in \mathrm{Aut}_A(\mathbb{U}_\Delta^<)$ and $g_{2i} \in \mathrm{Aut}_B(\mathbb{U}_\Delta^<)$. Roughly speaking, we start from an isometric copy of D in $\mathbb{U}_\Delta^<$ where all the new points (other than $A \cap B$) are appropriately distributed between x_is. Using the combination of g_is, we propagate the points in the initial copy of D to create the correct ordering of points in the final copy of D.

As a substructure of $\mathbb{U}_\Delta^<$, D is a finite ordered metric space. Define two order-preserving partial isometries p_1, p_2 of D by letting $p_1(x) = x$ for all $x \in A$ and $p_2(x) = x$ for all $x \in B$. Now applying Lemma 0.4 to D and p_1, p_2, we get a finite ordered metric space D' extending D, order-preserving partial isometries q_1, q_2 of D' which are extensions of p_1, p_2, respectively, and a word $w \in \mathsf{F}(s,t)$ such that

(1) $Z(q_1) = Z(p_1) = F(p_1) = A$ and $Z(q_2) = Z(p_2) = F(p_2) = B$;
(2) $D \subseteq \mathrm{dom}(q_1)$;
(3) $\mathrm{dom}(q_1) \cup w(q_1, q_2)(D)$ is the free amalgam of $\mathrm{dom}(q_1)$ and $w(q_1, q_2)(D)$ over $\{x_0\}$.

For notational simplicity we write $w(q_1, q_2)$ as w.

We note that $(A \cup B) \cap w(D) = \{x_0\}$. This is because $x_0 \in A \cup B \subseteq C \subseteq D \subseteq \mathrm{dom}(q_1)$ and by (3), $\mathrm{dom}(q_1) \cup w(D)$ is the free amalgam of $\mathrm{dom}(q_1)$ and $w(D)$ over $\{x_0\}$.

Let $w_1(D)$ be an isomorphic copy of $w(D)$ so that $w_1(D)$ and $w(D)$ have exactly one common point x_0. For each $y \in D$, the point in $w_1(D)$ corresponding to $w(y) \in w(D)$ will be denoted by $w_1(y)$. Let D_1' be the free amalgam of D' and $w_1(D)$ over $\{x_0\}$. Figure 1 illustrates the containment relationship among the sets considered so far.

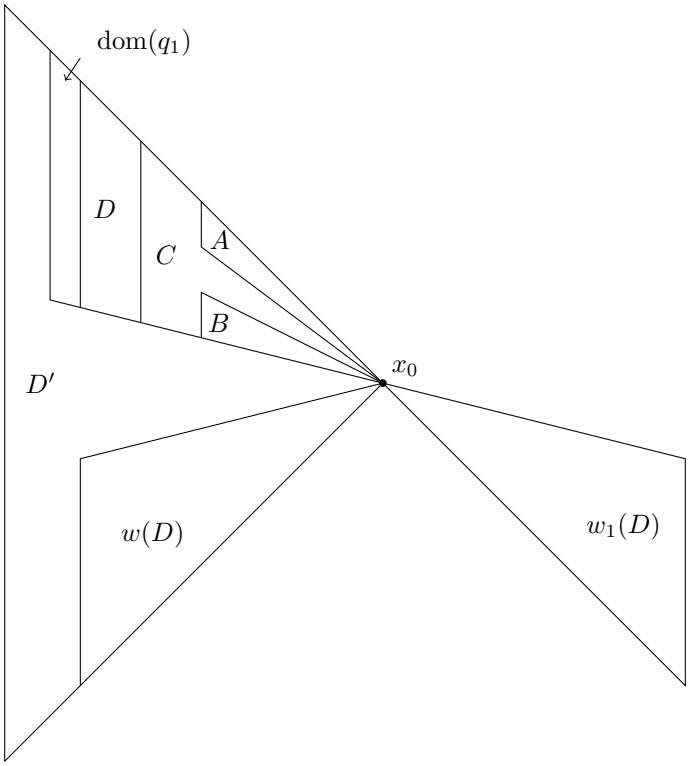


FIGURE 1. The composition of D_1'.

Extend the ordering on D' to D_1' so that the following hold:

(a) for any $y_1, y_2 \in D$, $w_1(y_1) < w_1(y_2)$ if and only if $w(y_1) < w(y_2)$;
(b) for any $u \in A \cup B$ and $y \in D$, if $w(y) < b_n = \max(A \cup B)$, then $u < w_1(y)$ if and only if $u < w(y)$;
(c) for any $y \in D$, if $w(y) > b_n$, then $a_n < w_1(y) < b_n$.

Such an extension clearly exists.

Now define $g_1 \colon A \cup w(D) \to A \cup w_1(D)$ by letting $g_1(a) = a$ for all $a \in A$ and $g_1(w(y)) = w_1(y)$ for all $y \in D$. Then g_1 is order-preserving by the above requirements (a) through (c). To see that g_1 is a partial isometry, let $a \in A$ and $y \in D$. Since $\mathrm{dom}(q_1) \cup w(D)$ is the free amalgam of $\mathrm{dom}(q_1)$ and $w(D)$ over $\{x_0\}$ and D_1' is the free amalgam of D' and $w_1(D)$ over $\{x_0\}$, we have

$$d(a, w(y)) = d(a, x_0) + d(x_0, w(y)) = d(a, x_0) + d(x_0, w_1(y)) = d(a, w_1(y)).$$

Intuitively, g_1 sends $w(y)$ to a copy which is smaller than b_n and fixes A pointwise. Consequently, we have $w_1(D) < b_n$ and, for any $y \in D$, $w_1(y) < x_0$ if and only if $w(y) < x_0$.

Let $w_2(D)$ be a fresh isomorphic copy of $w(D)$ such that $w_2(D)$ and $w(D)$ has exactly one common point x_0. Let D_2' be the free amalgam of D_1' and $w_2(D)$ over $\{x_0\}$. Extend the ordering on D_1' to D_2' so that the following hold:

(d) for any $y_1, y_2 \in D$, $w_2(y_1) < w_2(y_2)$ if and only if $w_1(y_1) < w_1(y_2)$;
(e) for any $u \in A \cup B$ and $y \in D$, if $w(y) < a_n$, then $u < w_2(y)$ if and only if $u < w_1(y)$;
(f) for any $y \in D$, if $w(y) > a_n$, then $b_{n-1} < w_2(y) < a_n$.

Define an order-preserving partial isometry $g_2 \colon B \cup w_1(D) \to B \cup w_2(D)$ by letting $g_2(b) = b$ for all $b \in B$ and $g_2(w_1(y)) = w_2(y)$ for all $y \in D$. Moreover, $w_2(D) < a_n$ and, for any $y \in D$, $w_2(y) < x_0$ if and only if $w_1(y) < x_0$.

Repeating this construction, we obtain, for $1 \leq i \leq 2n$, $D_i' = D' \cup w_1(D) \cup \cdots \cup w_i(D)$, which can be viewed as the free amalgam of $w_i(D)$ and the remaining part over $\{x_0\}$, an ordering on D_{2n}' extending that on D', and order-preserving partial isometries $g_1, g_2, g_3, \ldots, g_{2n}$ such that

(i) for any $1 \leq i \leq n$, $g_{2i-1}(a) = a$ for all $a \in A$, and $g_{2i}(b) = b$ for all $b \in B$;
(ii) for any $1 \leq i \leq n$ and $y \in D$, $g_{2i}(w_{2i-1}(y)) = w_{2i}(y)$, and $g_{2i-1}(w_{2i-2}(y)) = w_{2i-1}(y)$ if $i > 1$.

Moreover, we have $w_{2n}(D) < a_1$ and, for any $y \in D$, $w_{2n}(y) < x_0$ if and only if $w(y) < x_0$.

Define a map

$$r \colon A \cup w_{2n}(C) \to A \cup w_{2n}(p(C))$$

by letting $r(a) = a$ for all $a \in A$ and $r(w_{2n}(c)) = w_{2n}(p(c))$ for all $c \in C$. Then for $a \in A$ and $c \in C$, we have

$$\begin{aligned} d(a, w_{2n}(c)) &= d(a, x_0) + d(x_0, w_{2n}(c)) = d(a, x_0) + d(x_0, w(c)) \\ &= d(a, x_0) + d(x_0, c) = d(a, x_0) + d(x_0, p(c)) \\ &= d(a, x_0) + d(x_0, w(p(c))) \\ &= d(a, x_0) + d(x_0, w_{2n}(p(c))) = d(a, w_{2n}(p(c))) \\ &= d(r(a), r(w_{2n}(c))). \end{aligned}$$

Thus r is a partial isometry. Note that w and p are order-preserving and each fixes x_0. Thus for any $c \in C$, we have

$$\begin{aligned} w_{2n}(c) < x_0 &\iff w(c) < x_0 \iff c < x_0 \\ &\iff p(c) < x_0 \iff w(p(c)) < x_0 \\ &\iff w_{2n}(p(c)) < x_0. \end{aligned}$$

Since for all $y \in D$, $w_{2n}(y) < a_1$, we conclude that r is an order-preserving partial isometry.

By Lemma 0.4 and our construction, the distances in D'_{2n} are all taken from the additive semigroup generated by the distances in D. Now redefine the metric on D'_{2n} by taking the minimum of the original distance value and $\sup \Delta$. Call the resulting metric space D''. Since D is a Δ-metric space, it follows that D'' is a Δ-metric space, and, as an ordered metric space, D'' is an extension of D. Moreover, each of the order-preserving partial isometries $q_1, q_2, g_1, g_2, \ldots, g_{2n}, r$ continues to be order-preserving partial isometries of D''.

Now by the weak homogeneity of $\mathbb{U}^<_\Delta$, we can extend D to an isomorphic copy of D'' as a substructure of $\mathbb{U}^<_\Delta$. By the ultrahomogeneity of $\mathbb{U}^<_\Delta$, we can extend all the above order-preserving partial isometries to full automorphisms of $\mathbb{U}^<_\Delta$. Denote the resulting automorphisms by the same notation. Then for $1 \leq i \leq 2n$, we have $r, q_1, g_i \in \mathrm{Aut}_A(\mathbb{U}^<_\Delta)$ if i is odd, and $q_2, g_i \in \mathrm{Aut}_B(\mathbb{U}^<_\Delta)$ if i is even. Finally, for any $c \in C$,

$$p(c) = w(q_1, q_2)^{-1} g_1^{-1} \ldots g_{2n}^{-1} r g_{2n} \ldots g_1 w(q_1, q_2)(c).$$

Thus if we let

$$q = w(q_1, q_2)^{-1} g_1^{-1} \ldots g_{2n}^{-1} r g_{2n} \ldots g_1 w(q_1, q_2),$$

then $q \in \langle \mathrm{Aut}_A(\mathbb{U}^<_\Delta), \mathrm{Aut}_B(\mathbb{U}^<_\Delta) \rangle$ and $p \restriction C = q \restriction C$ as required. □

Lemma 0.6. *Let Δ be a countable distance value set, and let $A, B \subseteq \mathbb{U}^<_\Delta$ be finite subsets. Suppose $A \cap B = \varnothing$. Then*

$$\langle \mathrm{Aut}_A(\mathbb{U}^<_\Delta), \mathrm{Aut}_B(\mathbb{U}^<_\Delta) \rangle \textit{ is dense in } \mathrm{Aut}(\mathbb{U}^<_\Delta).$$

Proof. Let $p \in \mathrm{Aut}(\mathbb{U}^<_\Delta)$ and let $C \subseteq \mathbb{U}^<_\Delta$ be a finite subset. We show that there is a $q \in \langle \mathrm{Aut}_A(\mathbb{U}^<_\Delta), \mathrm{Aut}_B(\mathbb{U}^<_\Delta) \rangle$ such that $p \restriction C = q \restriction C$. Again we may assume

$A \cup B \subseteq C$. Consider the ordered Δ-metric space $C \cup p(C)$. Extend it to an ordered Δ-metric space $C \cup p(C) \cup \{x_0\}$, where the metric is defined by

$$d(x_0, y) = \operatorname{diam}(C \cup p(C)) = \max\{d(y_1, y_2) \colon y_1, y_2 \in C \cup p(C)\}$$

for any $y \in C \cup p(C)$, and the ordering is defined by $x_0 < y$ for all $y \in C \cup p(C)$. Such an extension can be found in $\mathbb{U}_\Delta^<$ by the weak homogeneity of $\mathbb{U}_\Delta^<$, and so we may assume $x_0 \in \mathbb{U}_\Delta^<$. Now consider $A' = A \cup \{x_0\}$, $B' = B \cup \{x_0\}$ and let $p' \in \operatorname{Aut}_{x_0}(\mathbb{U}_\Delta^<)$ be an extention of $p \restriction C \cup \{(x_0, x_0)\}$. Applying Lemma 0.5 to A', B' and p', we get a $q \in \langle \operatorname{Aut}_{A'}(\mathbb{U}_\Delta^<), \operatorname{Aut}_{B'}(\mathbb{U}_\Delta^<)\rangle$ such that $q \restriction C \cup \{x_0\} = p' \restriction C \cup \{x_0\}$. Then in particular $q \in \langle \operatorname{Aut}_A(\mathbb{U}_\Delta^<), \operatorname{Aut}_B(\mathbb{U}_\Delta^<)\rangle$ and $q \restriction C = p \restriction C$ as required. □

We also need the following two lemmas about proper open subgroups of $\operatorname{Aut}(\mathbb{U}_\Delta^<)$.

Lemma 0.7. *Let V be a proper open subgroup of $\operatorname{Aut}(\mathbb{U}_\Delta^<)$. Then there exists an element $x \in \mathbb{U}_\Delta^<$ such that $V \cdot x$ is finite.*

Proof. Toward a contradiction, assume $V \cdot x$ is infinite for all $x \in \mathbb{U}_\Delta^<$. We prove that V is dense. Since V is also closed, it follows that $V = \operatorname{Aut}(\mathbb{U}_\Delta^<)$, contradicting the properness of V.

Note that a basic open neighborhood of the identity element of $\operatorname{Aut}(\mathbb{U}_\Delta^<)$ has the form $\operatorname{Aut}_A(\mathbb{U}_\Delta^<)$, where A is a finite subset of $\mathbb{U}_\Delta^<$. Since V is an open subgroup, there is a finite $A \subseteq \mathbb{U}_\Delta^<$ such that $\operatorname{Aut}_A(\mathbb{U}_\Delta^<) \subseteq V$. By Neumann's lemma (see Corollary 4.2.2 of Hodges [2]), there exists $g \in V$ such that $g(A) \cap A = \varnothing$.

By Lemma 0.6, $\langle \operatorname{Aut}_A(\mathbb{U}_\Delta^<), \operatorname{Aut}_{g(A)}(\mathbb{U}_\Delta^<)\rangle$ is dense in $\operatorname{Aut}(\mathbb{U}_\Delta^<)$. This group is contained in V, and thus V is dense. □

Lemma 0.8. *Let V be a proper open subgroup of $\operatorname{Aut}(\mathbb{U}_\Delta^<)$. The following are equivalent:*

1. *There exists a unique $x \in \mathbb{U}_\Delta^<$ such that $V = \operatorname{Aut}_{\{x\}}(\mathbb{U}_\Delta^<)$.*
2. *V is maximal among proper closed subgroups of $\operatorname{Aut}(\mathbb{U}_\Delta^<)$.*

Proof. First suppose $V = \operatorname{Aut}_{\{x\}}(\mathbb{U}_\Delta^<)$. Assume H is a closed subgroup of $\operatorname{Aut}(\mathbb{U}_\Delta^<)$ such that $V \subsetneq H \subsetneq \operatorname{Aut}(\mathbb{U}_\Delta^<)$. Let $g \in H \setminus V$. Then $g(x) \neq x$ and $\operatorname{Aut}_{\{g(x)\}}(\mathbb{U}_\Delta^<) = g\operatorname{Aut}_{\{x\}}(\mathbb{U}_\Delta^<)g^{-1} = gVg^{-1} \leq H$. By Lemma 0.6, $\langle \operatorname{Aut}_{\{g(x)\}}(\mathbb{U}_\Delta^<), \operatorname{Aut}_{\{x\}}(\mathbb{U}_\Delta^<)\rangle$ is dense in $\operatorname{Aut}(\mathbb{U}_\Delta^<)$. Since it is a subgroup of H and H is closed, we have $H = \operatorname{Aut}(\mathbb{U}_\Delta^<)$, contradicting the properness of H.

For the converse, assume (2) holds. By Lemma 0.7, there exists an element $x \in \mathbb{U}_\Delta^<$ such that $V \cdot x$ is finite. We show that $V \subseteq \operatorname{Aut}_{\{x\}}(\mathbb{U}_\Delta^<)$. Otherwise, let $g \in V$ be such that $g(x) \neq x$. Then either $g(x) > x$ or $g(x) < x$. Since g is order-preserving, we get that either $g^{n+1}(x) > g^n(x)$ for all natural numbers n, or $g^{n+1}(x) < g^n(x)$ for all natural numbers n. In either case, $V \cdot x$ is infinite.

It is clear that $\operatorname{Aut}_{\{x\}}(\mathbb{U}_\Delta^<) \neq \operatorname{Aut}(\mathbb{U}_\Delta^<)$. Thus by the the maximality of V, we have $V = \operatorname{Aut}_{\{x\}}(\mathbb{U}_\Delta^<)$.

For the uniqueness, assume $\mathrm{Aut}_{\{x\}}(\mathbb{U}^<_\Delta) = \mathrm{Aut}_{\{y\}}(\mathbb{U}^<_\Delta)$ but $x \neq y$. Then by Lemma 0.6, $\langle \mathrm{Aut}_{\{x\}}(\mathbb{U}^<_\Delta), \mathrm{Aut}_{\{y\}}(\mathbb{U}^<_\Delta)\rangle$ is dense. But

$$\langle \mathrm{Aut}_{\{x\}}(\mathbb{U}^<_\Delta), \mathrm{Aut}_{\{y\}}(\mathbb{U}^<_\Delta)\rangle = \mathrm{Aut}_{\{x\}}(\mathbb{U}^<_\Delta) = \mathrm{Aut}_{\{y\}}(\mathbb{U}^<_\Delta),$$

and thus they are both the whole group, which is impossible. □

We are now ready to finish the proof of Theorem 0.3. Recall that the remaining direction is (i) implying (ii).

Let $f\colon \mathrm{Aut}(\mathbb{U}^<_\Delta) \to \mathrm{Aut}(\mathbb{U}^<_\Lambda)$ be a topological group isomorphism. By Lemma 0.8, for each $x \in \mathbb{U}^<_\Delta$, there is a unique $y \in \mathbb{U}^<_\Lambda$ such that $f(\mathrm{Aut}_{\{x\}}(\mathbb{U}^<_\Delta)) = \mathrm{Aut}_{\{y\}}(\mathbb{U}^<_\Lambda)$. We write $y = g(x)$. The map g is well-defined and bijective.

Note that for any $x, y, x', y' \in \mathbb{U}^<_\Delta$, $\mathrm{Aut}_{\{x,y\}}(\mathbb{U}^<_\Delta)$ and $\mathrm{Aut}_{\{x',y'\}}(\mathbb{U}^<_\Delta)$ are conjugate in $\mathrm{Aut}(\mathbb{U}^<_\Delta)$ if and only if $d(x,y) = d(x',y')$. Also note that

$$\mathrm{Aut}_{\{x,y\}}(\mathbb{U}^<_\Delta) = \mathrm{Aut}_{\{x\}}(\mathbb{U}^<_\Delta) \cap \mathrm{Aut}_{\{y\}}(\mathbb{U}^<_\Delta)$$

and similarly for $\mathrm{Aut}_{\{x',y'\}}(\mathbb{U}^<_\Delta)$. Thus by the definition of g, we have that for any $x, y, x', y' \in \mathbb{U}^<_\Delta$,

$$d(x,y) = d(x',y') \iff d(g(x),g(y)) = d(g(x'),g(y')).$$

Now for any $d \in \Delta$, we define $\theta(d) \in \Lambda$ by finding $x, y \in \mathbb{U}^<_\Delta$ such that $d(x,y) = d$, and then setting $\theta(d) = d(g(x),g(y))$. By the above observation, θ is well-defined and bijective. We show that θ witnesses the equivalence of Δ and Λ. Suppose (d_1,d_2,d_3) is a Δ-triangle. There exist $x, y, z \in \mathbb{U}^<_\Delta$ such that the distances among them are exactly (d_1,d_2,d_3). Then the distaces among $g(x), g(y), g(z)$ are $(\theta(d_1),\theta(d_2),\theta(d_3))$. Thus $(\theta(d_1),\theta(d_2),\theta(d_3))$ is a Λ-triangle. The converse also holds.

The proof of Theorem 0.3 is complete.

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